Project INSPIRE Design Document

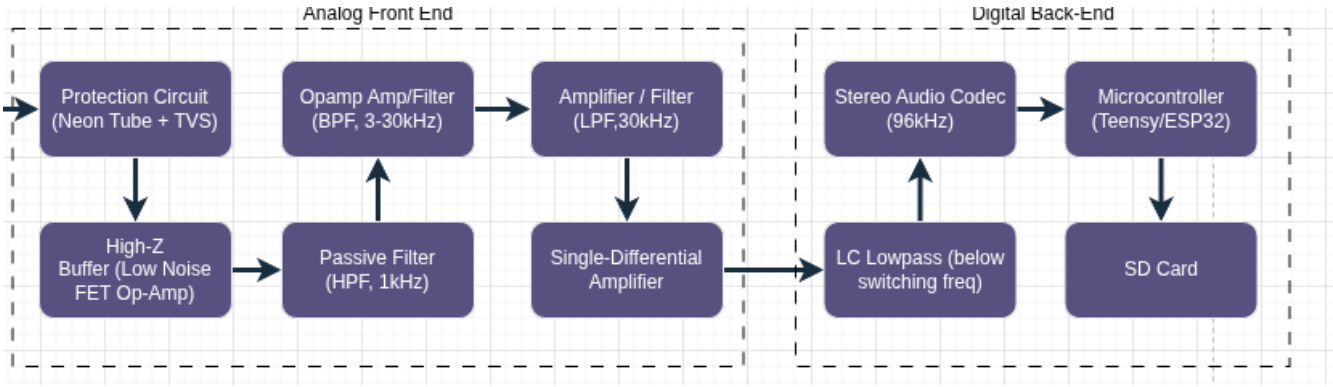
Project Part IV: Active Amplifier & Filter

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

The Opamp Amp/Filter acts as the primary active conditioning stage for the VLF receiver. It implements a active Band-Pass Filter (BPF) specifically tuned to isolate the 3-30kHz range. By introducing active gain, this block lets us amplify only the signals in the pass-band and provides further isolation from interference. The following design elements need to be accounted for

- Active Filter Topology: This schematic needs to use an active filter topology, like Sallen-Key or Multiple Feedback. These designs ensure a flat pass-band with a low-quality factor, given the need for the extreme wideband nature relative to frequency.
- Gain: This design will provide active gain. In the pass-band, 10dB of gain should be possible. This works in tandem with the other amplifiers and lets us ease the gain constraint off of the other amplifiers.
- Power Supply Constraints: Make sure that the op-amp selected for this design usage works on a 3.3V voltage rail.

3) Expected Outputs

This sub-circuit design requires a technical write-up and simulation covering three major parts:

- **Topology & Component Values:** A justification of filter topology (with appropriate schematic) and component values must be determined first. This includes existing cutoff values of 3kHz and 30kHz, the 10dB pass-band gain, and all relative resistor and capacitor values.
- **Op-Amp Selection:** Find a suitable operational amplifier, and give a justification for it, along with relevant parameters, like GBW, Slew Rate, and output impedance.
- **SPICE Simulation:** A two-stage SPICE simulation is required:
    - An AC analysis showing the filter response well past the region of interest. The rolloff on both sides needs to match

calculated values, and pass-band gain should match provided specification.

    - A Noise analysis needs to be run to ensure this element is not providing excess noise to the system.

Optional Extra (Stretch Goals):

- Run a DC transient analysis, and plot input and output. This tells you the linearity of your amplifier, which is important to prevent distortion.